Statistics for Machine Learning

A focused syllabus covering the statistical foundations needed to begin Machine Learning with scikit-learn. Advanced topics can be layered in later alongside real ML projects.

SYLLABUS • 5 CHAPTERS

Chapter 1 — Descriptive Statistics

- Mean
- Median
- Mode
- Range
- Quartiles
- Percentiles
- Variance
- Standard Deviation
- Interquartile Range (IQR)
- Outliers
- Skewness
- Kurtosis

Chapter 2 — Sampling

- Population vs Sample
- Sampling Techniques
- Sampling Bias
- Sampling Distribution
- Central Limit Theorem (CLT)

Chapter 3 — Inferential Statistics

- Confidence Intervals
- Standard Error
- Margin of Error
- Hypothesis Testing
- Null & Alternative Hypothesis
- p-value
- Significance Level
- Type I & Type II Errors

Chapter 4 — Correlation & Covariance

- Covariance
- Pearson Correlation
- Spearman Correlation
- Correlation Matrix
- Correlation vs Causation
- Multicollinearity (Overview)

Chapter 5 — Statistical Testing

- z-Test
- t-Test
- Chi-Square Test
- ANOVA
- Non-Parametric Tests (Overview)

Chapter 4 — Correlation & Covariance

A closer look at each topic in this chapter

Covariance

Measures how two variables change together — whether they move in the same direction (positive) or opposite directions (negative). Its value is unbounded and depends on the scale of the variables, which makes it hard to compare across datasets.

Pearson Correlation

A standardized measure (ranging from -1 to +1) of the linear relationship between two continuous variables. It's essentially covariance scaled by the standard deviations of both variables, making it easy to interpret and compare.

Spearman Correlation

A rank-based correlation measure that captures monotonic (not necessarily linear) relationships. It's more robust to outliers and works well when data is ordinal or doesn't follow a normal distribution.

Correlation Matrix

A table showing pairwise correlation coefficients between multiple variables at once. It's commonly used during exploratory data analysis to quickly spot strong relationships and potential redundancy among features.

Correlation vs Causation

Two variables can be strongly correlated without one causing the other — the relationship may be coincidental or driven by a hidden third variable (a confounder). Recognizing this distinction is essential before drawing conclusions from data.

Multicollinearity (Overview)

Occurs when two or more input features are highly correlated with each other, making it difficult for a model to isolate each feature's individual effect. This can destabilize coefficients in linear models and is usually checked using a correlation matrix or VIF (Variance Inflation Factor).

Note: This syllabus is sufficient to build the statistical foundation required to start Machine Learning with scikit-learn. Advanced topics can be learned later alongside ML projects.